

Intelligent Student Performance Analytics and Learning Strategy Optimization

A DATA MINING APPROACH USING THE OPEN UNIVERSITY LEARNING ANALYTICS DATASET

S.D. Prematunga
268382R

PROBLEM & OBJECTIVES

The Challenge

- Student dropout and failure remain persistent issues in higher education.
- Traditional identification of at-risk students is reactive and often too late.
- Need: Proactive, data-driven early-warning systems.

Objectives

- Identify key factors influencing student outcomes.
- Build predictive models to flag at-risk students early.
- Discover natural student segments through clustering.
- Generate actionable recommendations for educators

DATA SET & PREPARATION

Open University Learning Analytics Dataset (OULAD):

- **32,593** students across multiple course presentations
- **7** interrelated data files merged into a unified dataset

Data Integration Pipeline:

- **Student Assessments:** total_weighted_score, avg_score, assessments_submitted
- **VLE Interactions:** total_clicks, active_days, unique_activities
- **Registration Records:** earliest_registration, has_unregistered
- **Student Demographics:** gender, age_band, education, region, disability

Missing values are filled with 0

Engineered features (5):

- clicks_per_day, assessment_completion_rate, score_per_assessment, early_registration, is_at_risk

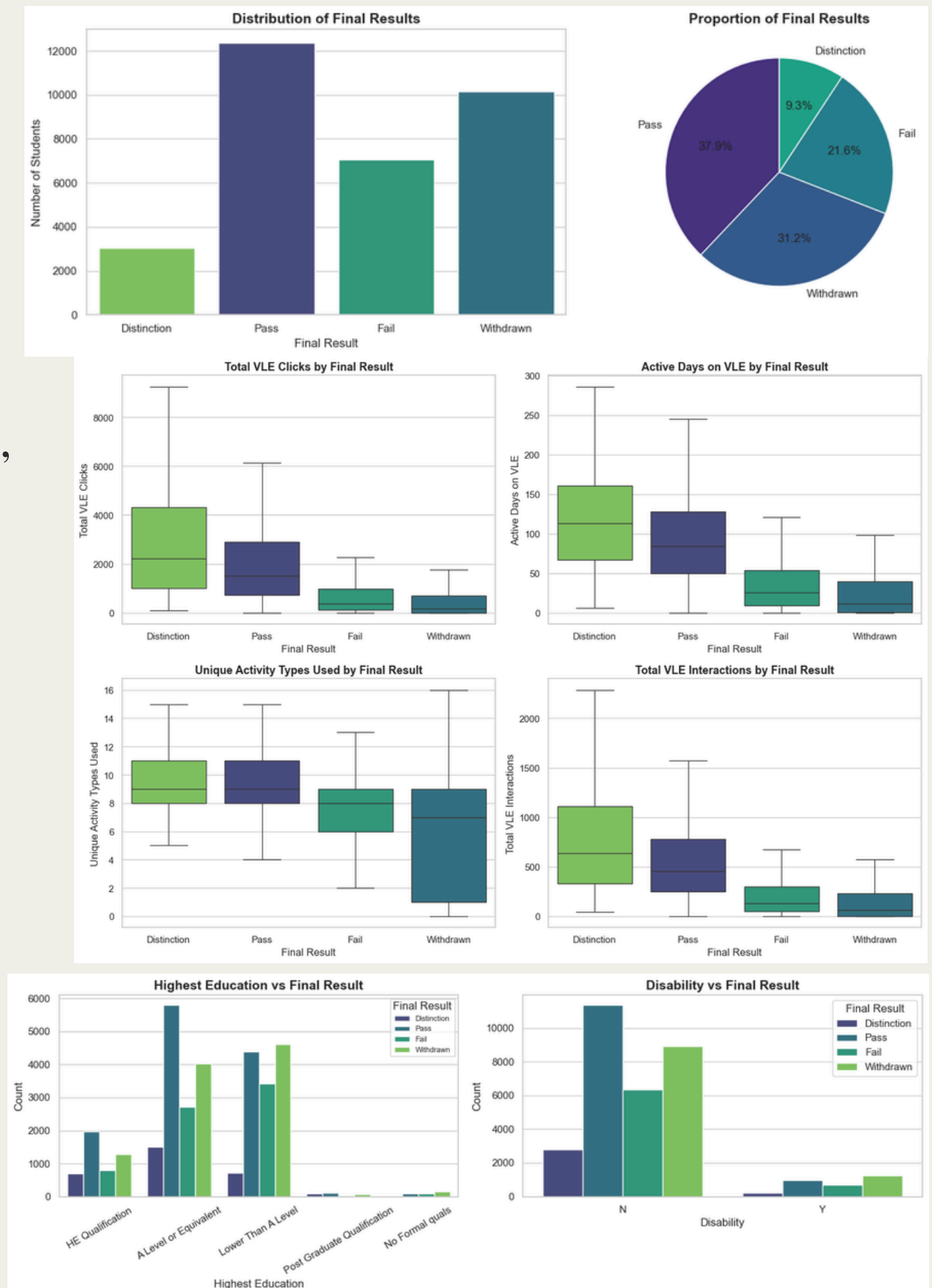
EXPLORATORY DATA ANALYSIS – KEY FINDINGS

Target Distribution:

- **52.8% At-Risk (Fail + Withdrawn) vs. 47.2% Successful**

Key Patterns Discovered:

- **Engagement** : Successful students exhibit substantially higher total clicks, active days, unique activities, and total VLE interactions.
- **Assessment performance**: Successful students exhibit Higher average score, higher total weighted score and higher number of assessment submissions.
- **Educational analysis**: Student has more prior qualifications exhibits a higher pass rate.
- **Disability**: Students who are disabled have a higher withdrawal rate.



PREDICTIVE MODELING RESULTS

Random Forest and Gradient Boosting Models with 80/20 data split. Assessment engagement dominates prediction. Students who submit and score well are far less likely to fail.

Metric	Value
Accuracy	0.9296
F1-Score	0.9319
ROC-AUC	0.9805
5-Fold CV Mean F1	0.9309

Table 1: Random Forest Results

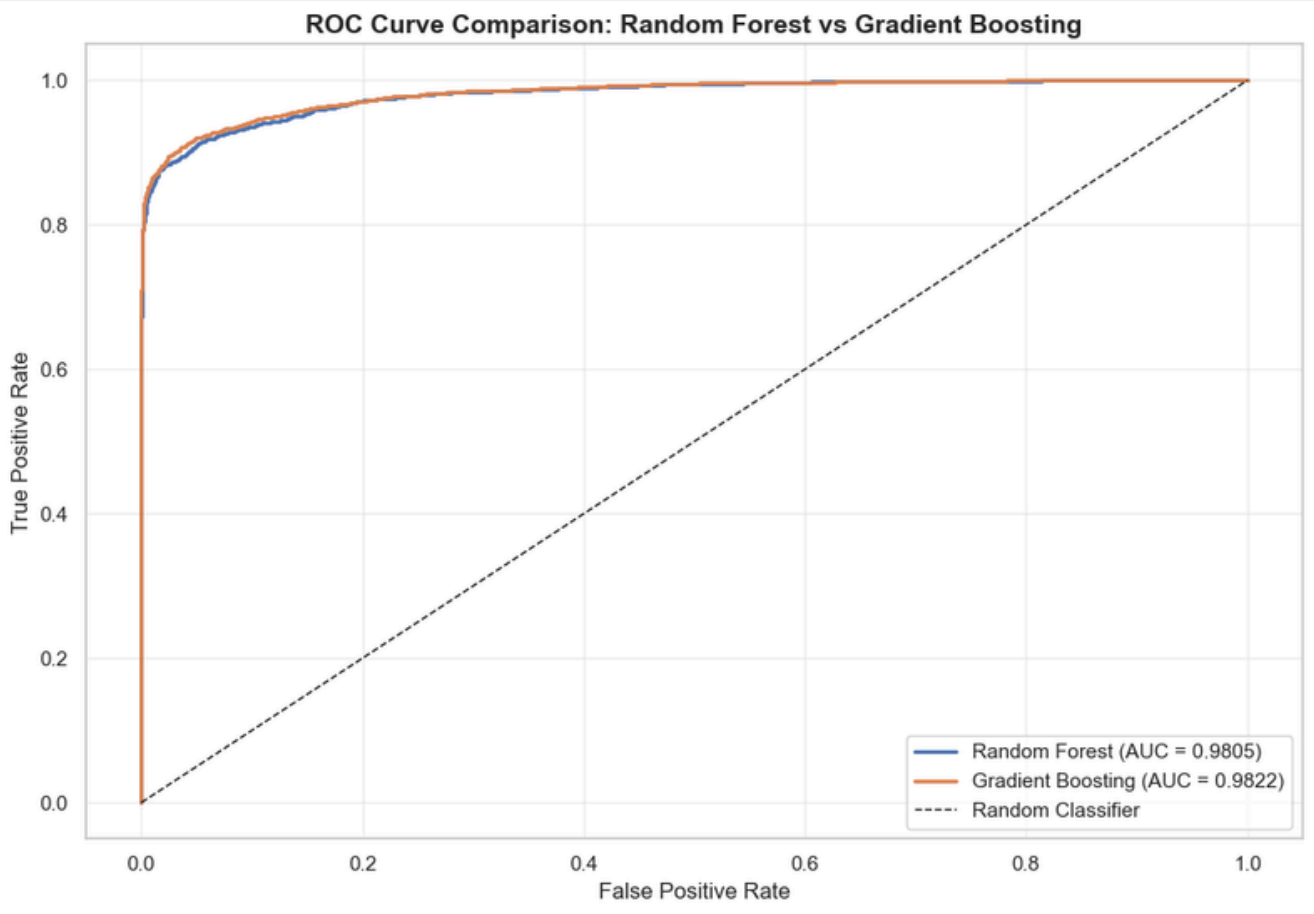
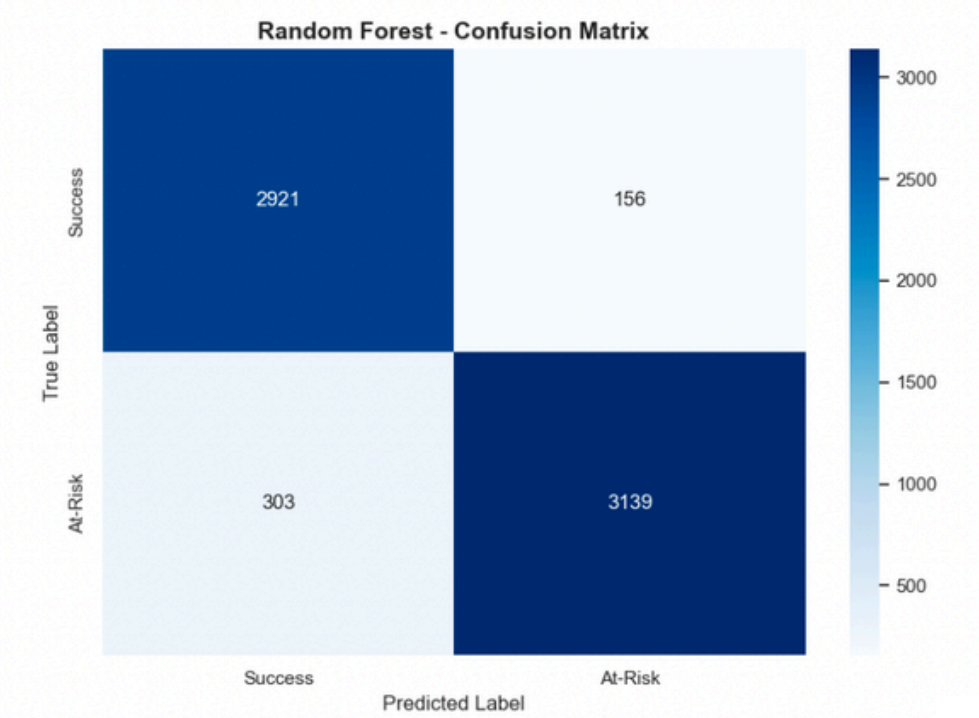


Fig. 7: Confusion matrix for the Random Forest classifier.

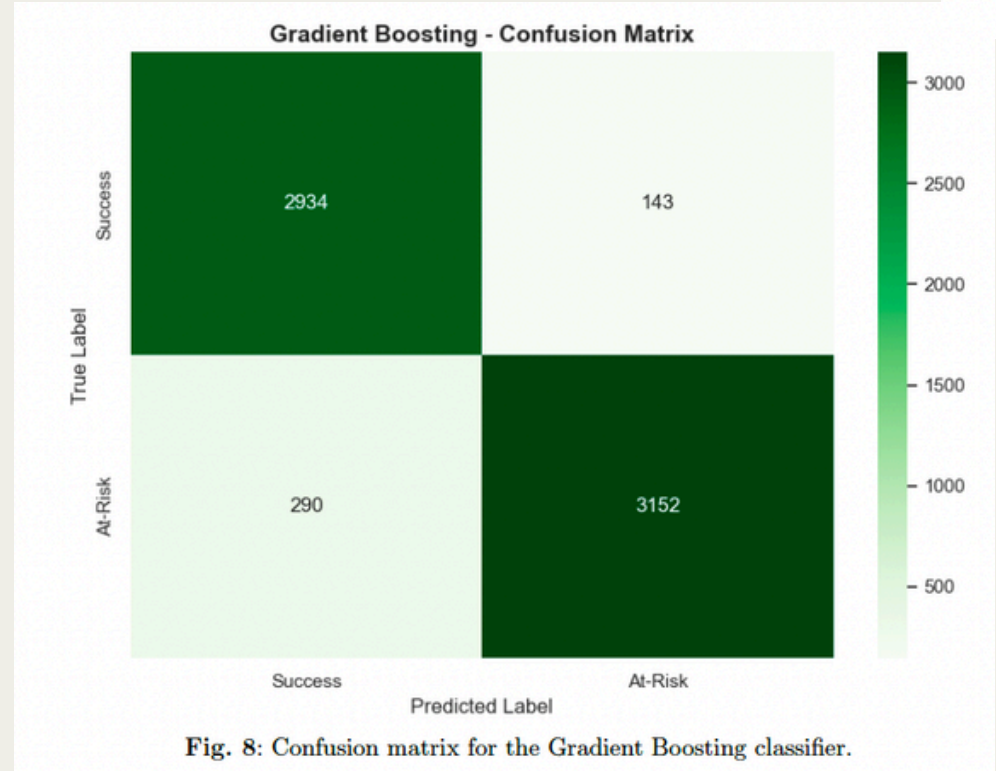
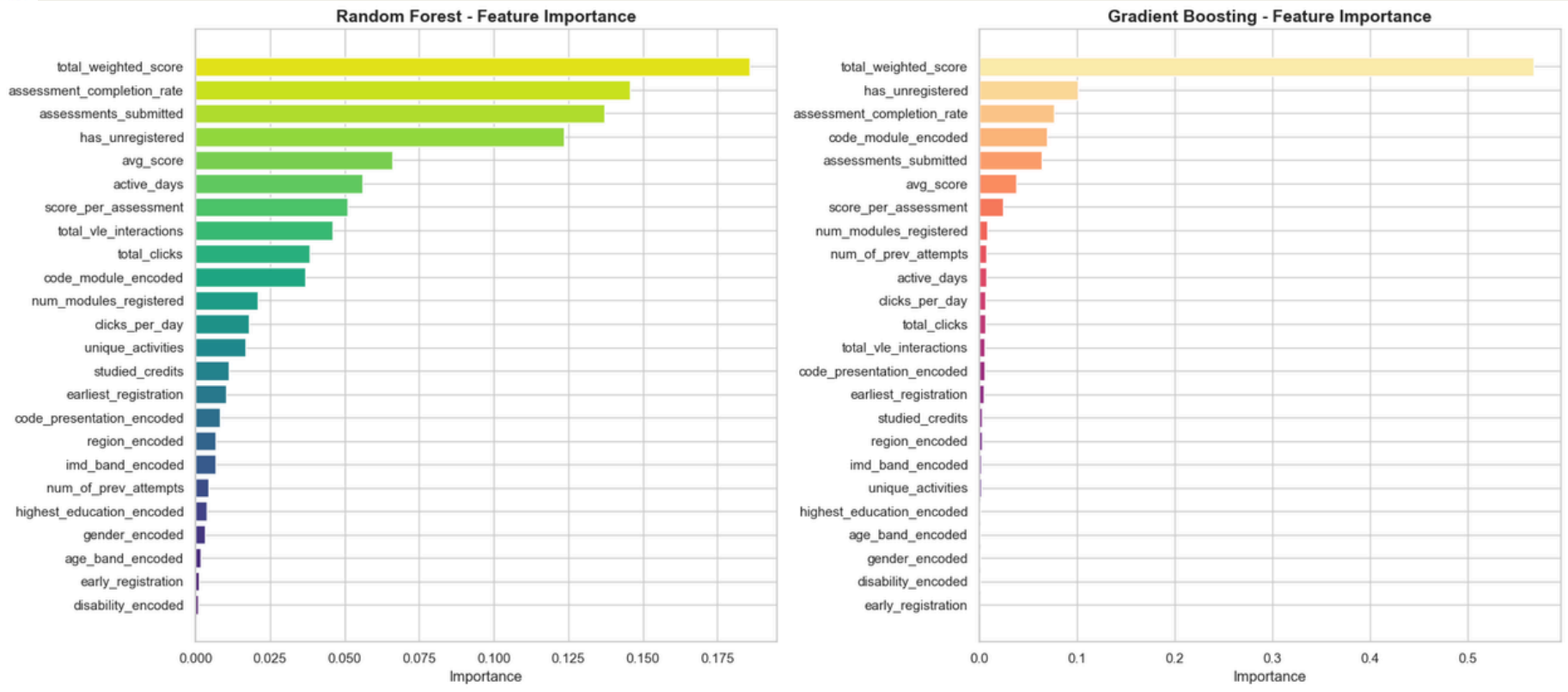


Fig. 8: Confusion matrix for the Gradient Boosting classifier.

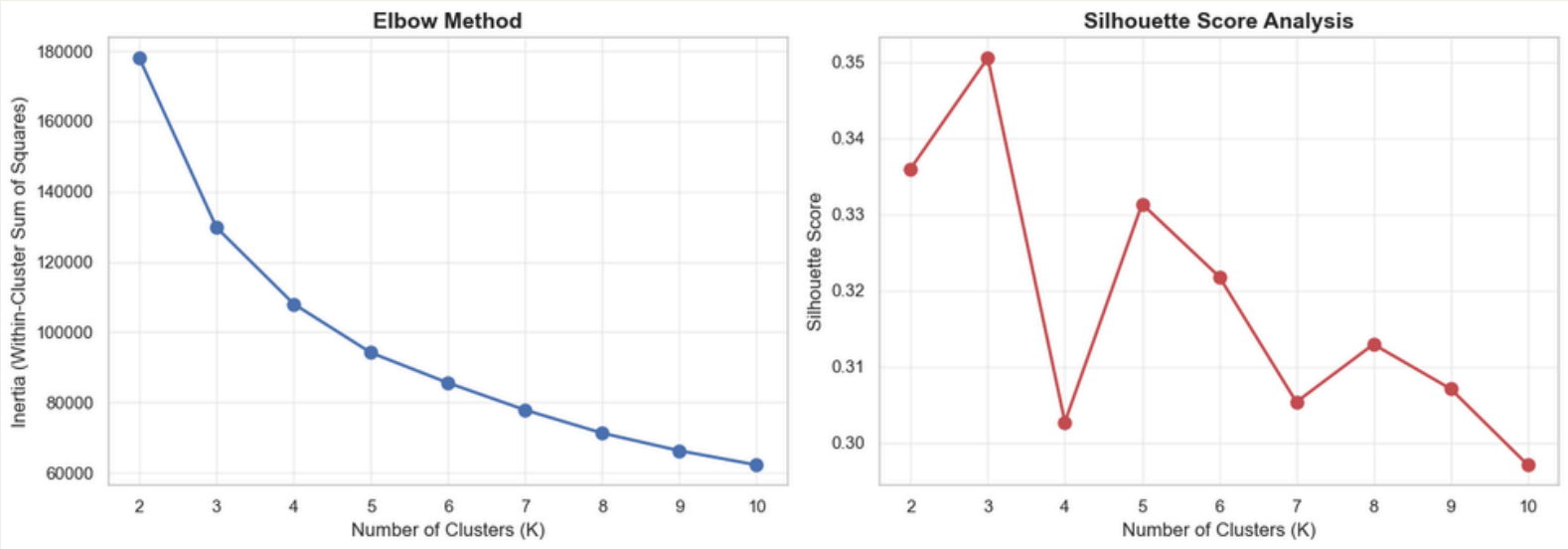
Metric	Value
Accuracy	0.9336
F1-Score	0.9357
ROC-AUC	0.9822
5-Fold CV Mean F1	0.9340

Table 2: Gradient Boosting Result



UNSUPERVISED LEARNING – STUDENT SEGMENTS

K-Means Clustering (K=3, Silhouette Score = 0.351):

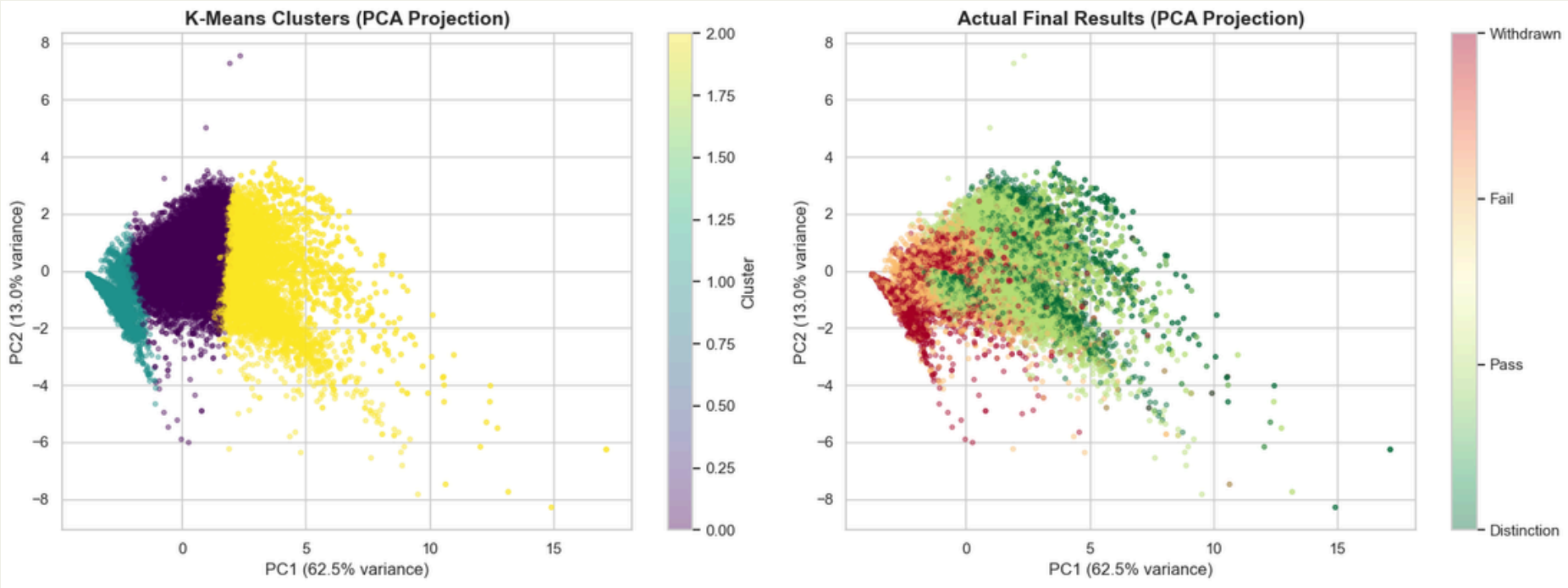


Feature	Cluster 0 (Moderate)	Cluster 1 (Disengaged)	Cluster 2 (High Performers)
Total Clicks	905.97	69.66	4239.72
Active Days	52.46	4.74	147.07
Avg Score	72.17	6.94	79.46
Total Weighted Score	54.18	0.69	133.89
Assessments Submitted	6.49	0.29	11.86

Table 3: Cluster profiles with mean values

Association Rule Mining:

- Apriori Algorithm: Min support 0.05, Confidence 0.5
- 7,320 Rules Discovered
- Key rule:
 - Low score + Inactive VLE + Few assessments $\Rightarrow$ Withdrawn (lift ~5.5)



RECOMMENDATIONS & CONCLUSION

Data-Driven Recommendations:

- Deploy early-warning system: Use a model to flag at-risk students in the early stage
- Prioritize Cluster 0 (Moderate) students: Engaged but at risk; highest intervention ROI
- Monitor module unregistration: Early red flag for advisor outreach
- Assessment completion support: Completion rate is a top predictor of success
- Investigate Cluster 1 barriers: Understand why disengaged students never participate

Ethical Considerations:

- Audit models for demographic bias
- Use predictions to support, not penalize students
- Ensure privacy compliance and student consent

Thank you!

S.D. Prematunga
268382R

Intelligent Student Performance Analytics and Learning Strategy Optimization: A Data Mining Approach

Shamil Prematunga

Contributing authors: shamildilshan.b@gmail.com;

Abstract

Predicting students at risk of academic failure and analyzing causes is critical when it comes to decision making stages. This study applies data mining techniques to the Open University Learning Analytics Dataset (OULAD) which contains around 32,000 data records with demographic, assessment and virtual learning environment (VLE) engagement data. This construct an analytical pipeline including data preparation, exploratory data analytics (EDA), supervised classifications and unsupervised learning. Under supervised classification this uses Random Forest and Gradient Boosting predictive models to predict at-risk students, achieving ROC-AUC scores of 0.9805 and 0.9822, respectively. Under unsupervised learning approach, K-Means clustering revealed three distinct student categories, while association rule mining (ARM) uncovers behavioral patterns in the dataset. The findings highlight that the performance related matrices such as total weighted score, assessment completion rate and submission counts are strong predictors in the analysis. The conclusion includes data driven recommendations for early warning system and academic support.

Keywords: Data Mining, Student Performance Analysis, Clustering, Association Rule Mining